Doubly Low-Resource? Evaluating Modern Language Models on Historical isiXhosa and the Role of Orthographic Variation

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Abstract

Historical isiXhosa combines limited NLP resources with variation across time. We compare XLM-R and AfroXLMR on historical newspapers and two contemporary corpora using word-normalised masked-language-model scores. Historical text has higher prediction loss; lexicon-guided normalisation reduces this loss while changing 5.83% of words. Historicisation also increases contemporary-text loss. These interventions support the relevance of spelling variation, but neither isolate the causes of the corpus gap nor justify treating historical spellings as errors.

Introduction

The isiXhosa Intellectual Traditions archive preserves newspapers documenting African intellectual, political and cultural life (Schoots, Ngwendu, De Wet, & Muftic, 2023). Computational access to these materials could support historical research, but models developed for contemporary language may respond differently to older spelling, vocabulary and contexts. We therefore characterise historical isiXhosa as a "doubly low-resource" setting, combining the limited NLP resources available for isiXhosa (Meyer & Buys, 2024) with the challenges posed by historical language variation (Bollmann, 2019). Distinguishing spelling-related difficulty from broader corpus differences matters both for evaluating models and for deciding whether normalisation is a useful, culturally responsible intervention.

We ask: **RQ1**, how does historical isiXhosa prediction difficulty compare with contemporary isiXhosa, and where is it concentrated? **RQ2**, how much does historical spelling variation contribute to the measured gap? We contribute two-model, two-reference comparisons, word-category decomposition and regression analyses, alongside tests of frozen spelling transformations and a random-noise control.

Background and Related Work

Historical normalisation maps spelling variants to modern forms; Bollmann (2019) compares rule-based, statistical and neural approaches and shows why evaluation depends on data and method. We use a conservative dictionary rather than assume every historical variant is an error. XLM-R is a multilingual masked language model (Conneau et al., 2020); AfroXLMR adapts multilingual representations to African languages (Alabi, Adelani, Mosbach, & Klakow, 2022). Their comparison probes whether adaptation's observed advantages extend equally to historical text. Pseudo-log-likelihood enables evaluation without task-specific training (Salazar, Liang, Nguyen, & Kirchhoff, 2020); the within-word masking refinement of Kauf and Ivanova (2023) motivates our scoring method.

Methodology

Data. Archive text from *Isigidimi sama-Xosa* and *Imvo Zabantsundu* was cleaned, deduplicated, language-filtered and divided into non-overlapping 32-whitespace-word chunks within blocks, dropping remainders. Historical reconstruction reused frozen edition-level train/development/test assignments (target 60/20/20); unmatched editions were excluded. Test sampling capped chunks per edition at 20, yielding 1,780 (780 *Isigidimi*; 1,000 *Imvo*). Contemporary references are Vuk'uzenzele news (Lastrucci, Rajab, Shingange, Njini, & Marivate, 2023) (M1, 135 chunks) and MzansiText (Lombard et al., 2026) (M2, 999, after excluding one corrupt chunk across conditions). The 75,254-type lexicon uses Vuk'uzenzele train/eval and MzansiText validation text, with frequency ≥ 2. Historical training supported discovery, development supported selection, and test data supported evaluation.

Scoring. We evaluate the base models without task-specific fine-tuning. Building on pseudo-log-likelihood scoring (Salazar et al., 2020), PLL-word-l2r (Kauf & Ivanova, 2023) masks the target subword and all later subwords of the same word, preventing within-word rightward leakage. For chunk c with W_c words and S_c scored subwords, we use

$$\ell_c = -\frac{1}{W_c} \sum_{i=1}^{S_c} \log P_\theta(t_i \mid x_c^{(M_i)}), \tag{1}$$

where M_i is that mask set. This pseudo-negative log-likelihood (NLL) is measured in nats/word; lower is better. Transformed chunks retain their original word-count denominator.

Interventions. Edit-distance matching of historical training out-of-vocabulary (OOV) words to the modern lexicon generated rule candidates. Positive pointwise mutual information contexts from training/non-test text compared candidate similarity with same-length, frequency-matched controls. Validation (2,000 bootstrap draws; positive lower 95% bound; at least 20 pairs) accepted aspiration-related (ASP) and $b \to bh$ (BH) families. Development-set OOV reduction selected parameters before freezing the lexicon-guided dictionary.

We applied the dictionary to historical test text (*normalisation*), inverse mappings at development-estimated family rates to contemporary text (*historicisation*), and random letter edits at its observed word-change rate (*noise*). Realised M1 change rates are 5.72% (historicisation) and 4.95% (noise); matching is approximate and does not equate edit types or plausibility.

Analysis. We define gap $G = \bar{\ell}_H - \bar{\ell}_M$, benefit $\delta = \bar{\ell}_H - \bar{\ell}_{H\text{-norm}}$ and gap closure $GC = 100\delta/G$ (%) using mean chunk loss. Contemporary costs are transformed minus original loss. We used 10,000 edition/document bootstrap draws for percentile 95% CIs, retaining paired interventions and joint nu-

merator/denominator draws for GC (seed 13). Approximate bootstrap tail areas for gaps and cluster sign-flip tests for paired effects entered Holm correction (Holm, 1979): four gaps, two benefits and two M1 historicisation effects. Categories are mutually exclusive in priority order: number/punctuation, English-list, capitalised, rewritten, OOV, in-lexicon. Word-level regression controls $\log_2$ subword count and lexical indicators, with cluster-robust CIs.

Results

Historical–Modern Performance Gap

Historical loss exceeds both references for both models (Table 1); all gap CIs are positive. AfroXLMR nevertheless has lower absolute historical NLL (11.576 versus XLM-R's 14.247), despite its larger gaps. Gap size also depends on the contemporary reference.

Table 1: Historical–modern gaps G (nats/word) and gap closure GC (%), with 95% cluster-bootstrap CIs.

Reference	G	GC
XLM-R		
M1	0.804 [0.447, 1.182]	38.2 [25.9, 67.7]
M2	1.851 [1.582, 2.107]	16.6 [14.4, 19.4]
AfroXLMR		
M1	2.887 [2.603, 3.174]	13.8 [12.4, 15.5]
M2	3.367 [3.163, 3.569]	11.9 [10.9, 12.9]

For XLM-R/M1, capitalised, OOV and rewritten categories contribute 2.009, 1.053 and 0.886 nats/word, offset by in-lexicon words (-2.996) and the remaining categories (-0.148). These share-weighted contributions sum to G; they are not within-category loss differences. The adjusted historical regression coefficient is 1.353 [1.038, 1.668], up from 0.804: the included controls do not remove this association.

Orthographic Intervention Effects

Normalisation reduces NLL by 0.307 [0.285, 0.330] for XLM-R and 0.400 [0.374, 0.425] for AfroXLMR. Changing 5.83% of historical words closes 11.9–38.2% of the gaps. M1 historicisation costs are 0.444 [0.367, 0.522] and 0.649 [0.554, 0.744]; both benefits and M1 costs have Holm-adjusted $p < 0.001$. Noise costs are descriptively larger (0.642 and 0.804; Figure 1). M2 also shows larger noise costs (0.568, 0.771) than historicisation costs (0.418, 0.541). No direct significance test compares these interventions.

Discussion

These comparisons identify a corpus-specific prediction penalty, not an intrinsic deficiency of historical isiXhosa. AfroXLMR's lower absolute loss but larger gap suggests uneven transfer of African-language adaptation benefits across periods (Alabi et al., 2022), although adaptation is not isolated causally. Capitalised and OOV contributions combine category prevalence and loss, identifying targets for further investigation. The adjusted coefficient indicates an association after the chosen controls.

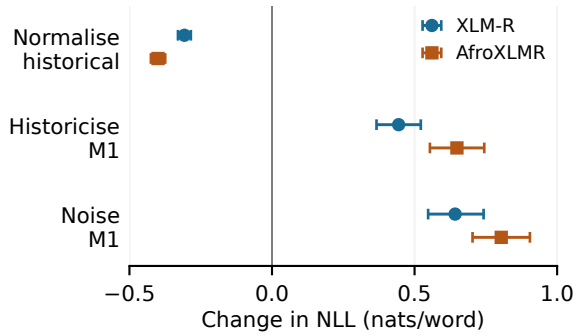


Figure 1: Changes in NLL (after minus before), with 95% paired cluster-bootstrap CIs. Modern interventions use M1; interval overlap is not a direct test of their difference.

Within-text spelling interventions support the practical relevance of historical normalisation (Bollmann, 2019). Gap closure measures this normaliser's benefit, not spelling variation's total causal contribution. Reference dependence and limited coverage leave room for uncovered orthography, genre, topic and other linguistic differences. Noise's larger point estimates suggest broader sensitivity to character changes, but approximate rate matching and the absence of a direct test prevent claims of a significant contrast or uniquely disruptive historical spelling.

Limitations include genre, period and topic differences; digitisation and filtering biases; missing editions in historical reconstruction; and only two validated rule families without exhaustive human checks. Block remainders and corrupt-input exclusion also restrict coverage. Pseudo-NLL measures predictability, not comprehension or downstream usefulness. Specialist validation and genre-matched evaluation are needed.

Historical spellings carry cultural meaning within an archive of African intellectual expression (Schoots et al., 2023). Normalisation risks obscuring that variation. Tools should improve access while preserving originals, exposing rewrites and enabling verification by isiXhosa speakers, language specialists and historians.

Conclusion

For RQ1, both models find the sampled historical texts less predictable than either modern reference; the gap varies by word category and persists after the included regression adjustments. For RQ2, the tested normaliser reduces loss and closes 11.9–38.2% of the gap, while historicisation increases modern-text loss. Orthography therefore matters for these model scores, but the experiments do not quantify its total causal contribution.

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Contribution Statement

The project was organised into three primary workstreams. Jordan led the foundational implementation, including data collection, preprocessing, corpus preparation, dataset splitting, and the initial experimental pipeline. Okwukwe extended this foundation by developing and evaluating the orthographic interventions, language-model scoring procedures, and statistical analyses used to investigate the research questions. Mivuyo coordinated the compilation and presentation of the research report, integrating the methodology, experimental findings, figures, and references into the final manuscript.

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If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2430639 (filled in on the group's behalf)

Candidate signature: 

Date: 2026/11/02